Multimedia Signal Processing - 2nd Module

August, 30 2021

1. **Question (8 pts)** Consider two sources X and Y that emit symbols according to the following joint p.m.f.

	x_1	x_2	x_3
y_1	1/8	1/8	0
y_2	0	1/4	0
y_3	0	1/8	1/4
y_4	0	1/8	0

   (a) Compute the probability mass functions of symbols of X and Y, separately.

   (b) Compute the entropy of both sources, i.e., $H(X)$ and $H(Y)$.

   (c) Compute the joint entropy $H(X, Y)$.

   (d) Design the Huffman code to encode single symbols of X and compare the average code length to $H(X)$.

   (e) Design the Huffman code to jointly encode symbols of X and Y and compare the average code length to $H(X, Y)$.

2. **IF ONLINE: Question (7 pts)**
   Oral examination.

3. **IF IN CLASS: Question 4 pts)** Explain the working principle of a typical video encoder based on a mixed DPCM-transform coding strategy. Draw a block diagram, and explain all input-output relationships, highlighting the role of motion vectors.

4. **IF IN CLASS: Question (3 pts)** Describe the concept of absolute threshold of hearing and explain how it relates to masking effects. How is it possible to use these concepts in perceptual audio coding?

5. **Questions (8 pts - each answer can be either TRUE or FALSE - Incorrect answers give you -1 points, unanswered 0 points, correct answers 1 point - Provide a brief comment when in doubt)**

- [T] [F] Given two independent discrete sources X and Y, mutual information $I(X;Y)$ must be equal to zero.
- [T] [F] If a source is characterized by a finite elements alphabet, it is possible to apply both lossy and lossless coding schemes.
- [T] [F] In image coding schemes, luminance and chrominance components are always given the same importance.
- [T] [F] Lloyd-Max quantizer provides the same performance of a uniform scalar quantizer if applied to a uniformly distributed source.
- [T] [F] The spectral envelop of voiced signals is characterized by a set of so-called formants.
- [T] [F] In a transform coding scheme, the used of KLT may introduce additional header information compared to the use of DCT.
- [T] [F] Any kind of square matrix can be used for transform coding.
- [T] [F] DPCM may provide more distortion than PCM at the same bitrate.

6. **Question (Matlab - 10pts)**

Load the default Matlab image `peppers.png` using the command `imread('peppers.png')`.

(a) Display the image.

(b) Extract the red color plane.

(c) Considering the red image component, split it into 4×4 non-overlapping blocks.

(d) Apply 2D DCT to each block.

(e) Compute and display the histograms (across blocks) of the DC coefficients and one AC coefficient of you choice.

(f) Set all the AC coefficients equal to zero for all blocks, thus only keeping the DC coefficient.

(g) Apply inverse 2D DCT to the modified blocks.

(h) Put the blocks together to recreate an image the same size of the original red component.

(i) How do you expect this image should look like compared to the original red component?